

AI Powered Debt Relief & Financial Recovery Platform

Project Description

The AI Powered Debt Relief & Financial Recovery Platform is an intelligent web application designed to simplify debt management and assist borrowers in recovering from financial difficulties. The platform leverages Artificial Intelligence to analyze a user's financial condition, calculate debt repayment capacity, recommend personalized repayment strategies, and generate AI-powered settlement and negotiation plans.

The system enables users to securely manage loan information, monitor repayment progress, evaluate financial health, and receive customized financial guidance. By integrating Google Gemini AI with modern web technologies such as React.js, FastAPI, Python, SQLite, and SQLAlchemy ORM, the application provides real-time intelligent recommendations through a responsive and user-friendly interface.

The platform aims to reduce financial stress, improve repayment planning, and support users in achieving long-term financial stability.

Scenarios

Scenario 1

Debt Analysis

A borrower logs into the platform and enters information about multiple loans, monthly income, and expenses. The AI analyzes the financial situation and calculates a realistic repayment plan based on disposable income.

Scenario 2

AI Negotiation Strategy

A user struggling to pay a personal loan requests assistance. The AI generates a professional negotiation strategy suggesting an affordable settlement amount and repayment schedule that can be discussed with the lender.

Scenario 3

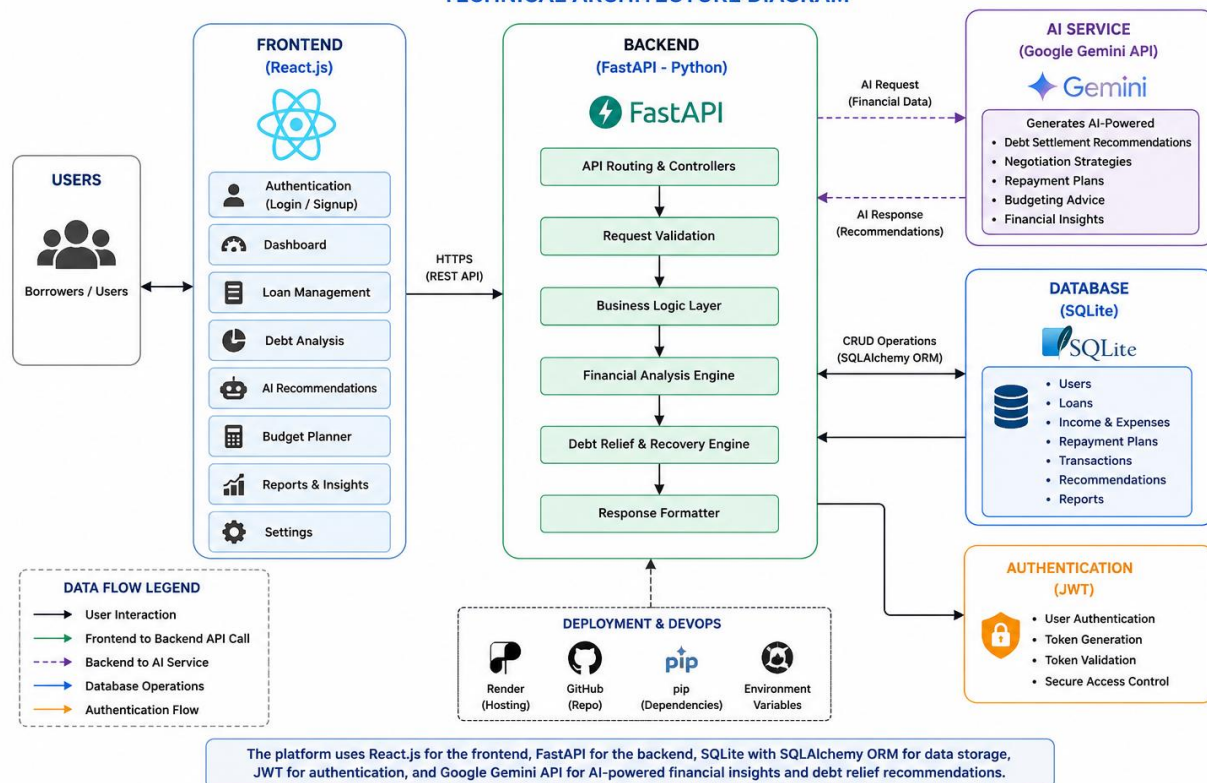
Financial Recovery Plan

After reviewing the user's financial profile, the AI creates a personalized recovery roadmap including budgeting advice, debt prioritization, and monthly savings recommendations to help the user become debt-free.

Technical Architecture:

AI POWERED DEBT RELIEF & FINANCIAL RECOVERY PLATFORM

TECHNICAL ARCHITECTURE DIAGRAM



Description

The **AI Powered Debt Relief & Financial Recovery Platform** follows a simple client-server architecture. Users access the application through a **React.js** web interface where they can register, log in, add loan details, income, and expenses. The frontend sends user requests to the **FastAPI** backend using REST APIs.

The backend processes the requests, validates the data, and stores it securely in the **SQLite** database using **SQLAlchemy ORM**. When a user requests financial advice or debt settlement recommendations, the backend sends the required financial information to the **Google Gemini API**. The AI analyzes the data and generates personalized repayment plans, budgeting advice, and debt negotiation strategies.

The backend receives the AI response and sends it back to the frontend, where it is displayed on the user's dashboard. **JWT Authentication** is used to provide secure login and protect user information.

Pre-requisites

- Python Programming Knowledge
- React.js Fundamentals
- FastAPI Framework Knowledge
- Google Gemini API Integration
- SQLite Database Knowledge
- SQLAlchemy ORM
- JWT Authentication
- HTML, CSS, and JavaScript
- REST API Concepts
- Git & GitHub Version Control
- VS Code Development Environment
- Render Deployment Knowledg

Project Workflow:

Milestone 1: Requirement Analysis and Planning

Activity 1.1: Identify the financial challenges faced by borrowers and define the project objectives.

Activity 1.2: Analyze the requirements for an AI-powered debt management and financial recovery platform.

Activity 1.3: Select Google Gemini AI as the AI model for generating financial recommendations and debt negotiation strategies.

Activity 1.4: Design the overall system architecture and finalize the technology stack (React.js, FastAPI, Python, SQLite, SQLAlchemy ORM, JWT).

Milestone 2: Backend Development

Activity 2.1: Develop the FastAPI backend to process user requests and business logic.

Activity 2.2: Create the SQLite database using SQLAlchemy ORM for storing user and loan information.

Activity 2.3: Integrate JWT authentication for secure user registration and login.

Activity 2.4: Connect the backend with Google Gemini API to generate AI-powered financial insights and recommendations.

Milestone 3: Core Feature Development

Activity 3.1: Develop the Loan Management module for adding, updating, and managing loan details.

Activity 3.2: Implement the Financial Health Analysis module to calculate debt-to-income ratio and repayment capacity.

Activity 3.3: Develop the AI Debt Settlement Recommendation module using Google Gemini AI.

Activity 3.4: Create the AI Negotiation Strategy Generator to provide personalized debt negotiation suggestions.

Activity 3.5: Develop the Budget Planner to recommend monthly budgets and savings plans.

Milestone 4: Frontend Development

Activity 4.1: Design a responsive user interface using React.js.

Activity 4.2: Develop pages for user registration, login, dashboard, and loan management.

Activity 4.3: Connect the frontend with FastAPI REST APIs.

Activity 4.4: Display AI-generated recommendations, repayment plans, and financial reports on the dashboard.

Milestone 5: Testing and Deployment

Activity 5.1: Perform functional testing of all application modules.

Activity 5.2: Test database operations, API integration, and user authentication.

Activity 5.3: Fix bugs and optimize application performance.

Activity 5.4: Deploy the application using Render and verify all features.

Milestone 6: Project Completion and Documentation

Activity 6.1: Prepare the complete project documentation.

Activity 6.2: Upload the source code to GitHub.

Activity 6.3: Demonstrate the application and submit the final project report and presentation.

Milestone 1: Requirement Gathering and Analysis

The first milestone of the AI Powered Debt Relief & Financial Recovery Platform focuses on understanding the financial challenges faced by borrowers and defining the project's objectives. During this phase, the system requirements, AI capabilities, architecture, and technology stack are carefully planned to ensure the platform provides intelligent debt management and financial recovery solutions.

Activity 1.1: Identify Financial Challenges and Define Project Objectives

Description

The initial step is to understand the common problems faced by borrowers, such as managing multiple loans, high debt, lack of financial planning, and difficulty negotiating with lenders. Based on these challenges, the project objectives are defined to develop an AI-powered platform that helps users analyze their financial health, manage debts efficiently, and receive personalized repayment and settlement recommendations.

Activity 1.2: Analyze System Requirements

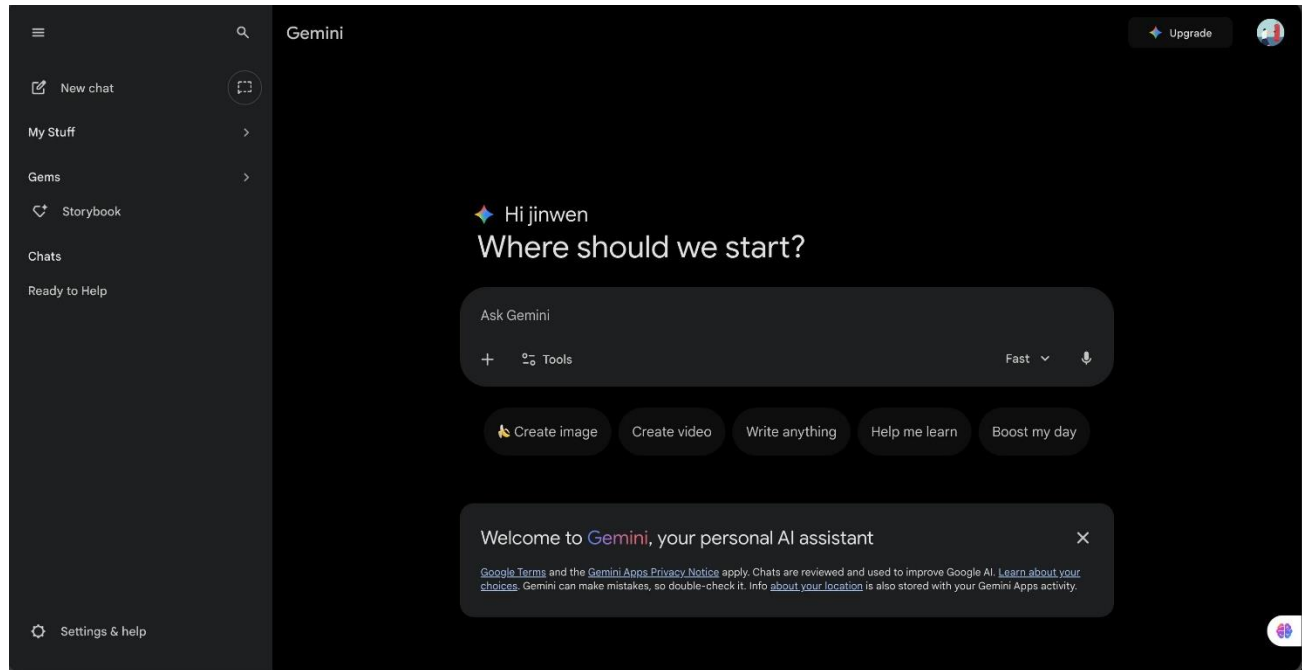
Description

The functional and non-functional requirements of the platform are analyzed. Functional requirements include user registration, loan management, debt analysis, AI recommendations, budgeting, and report generation. Non-functional requirements focus on security, performance, scalability, usability, and data privacy. This analysis ensures the platform meets user needs and supports future enhancements.

Activity 1.3: Select Google Gemini AI Model

Description

Google Gemini AI is selected as the core AI model because it can analyze financial information and generate intelligent debt settlement recommendations, budgeting advice, negotiation strategies, and personalized financial recovery plans. The AI improves decision-making by providing customized suggestions based on each user's financial condition.



Activity 1.4: Design System Architecture and Technology Stack Description

The overall architecture of the platform is designed using a modern client-server model. The frontend is developed with **React.js**, the backend with **FastAPI (Python)**, data is stored in **SQLite** using **SQLAlchemy ORM**, user authentication is secured using **JWT**, and **Google Gemini API** provides AI-powered financial recommendations. GitHub is used for version control, and the application is deployed using Render.

Milestone 2: Backend Development

This milestone focuses on developing the backend of the **AI Powered Debt Relief & Financial Recovery Platform** using **FastAPI** and **Python**. The backend is responsible for handling user requests, managing business logic, communicating with the database, and integrating the Google Gemini AI API to provide intelligent debt management and financial recovery recommendations.

Activity 2.1: Develop the FastAPI Backend

Description

The FastAPI backend is developed to process user requests efficiently. It provides REST APIs for user registration, authentication, loan management, financial analysis, AI recommendation generation, and report retrieval. FastAPI ensures high performance, automatic API documentation, and secure communication between the frontend and backend.

```
Code Blame 12 lines (10 loc) · 428 Bytes

1  from sqlalchemy import Column, Integer, String, Float
2  from database import Base
3
4  class Loan(Base):
5      __tablename__ = "loans"
6
7      id = Column(Integer, primary_key=True, index=True)
8      loan_name = Column(String, nullable=False)
9      outstanding_amount = Column(Float, nullable=False)
10     emi = Column(Float, nullable=False)
11     monthly_income = Column(Float, nullable=False)
12     overdue_months = Column(Integer, nullable=False)
```

```
1  from sqlalchemy import create_engine
2  from sqlalchemy.orm import declarative_base, sessionmaker
3
4  DATABASE_URL = "sqlite:///./debt_relief.db"
5
6  engine = create_engine(
7      DATABASE_URL,
8      connect_args={"check_same_thread": False}
9  )
10
11  SessionLocal = sessionmaker(
12      autocommit=False,
13      autoflush=False,
14      bind=engine
15  )
16
17  Base = declarative_base()
18  def get_db():
19      db = SessionLocal()
20      try:
21          yield db
22      finally:
23          db.close()
```

Activity 2.2: Create the SQLite Database Using SQLAlchemy ORM

Description

The SQLite database is designed to securely store user information, loan details, income records, expenses, repayment history, and AI-generated recommendations. SQLAlchemy ORM is used to simplify database operations such as creating tables, inserting records, updating data, and retrieving information efficiently.

Activity 2.3: Implement JWT Authentication

Description

JWT (JSON Web Token) authentication is implemented to provide secure user registration and login. After successful authentication, a secure token is generated and used to authorize users when accessing protected features such as loan management, financial analysis, and AI recommendations.

Activity 2.4: Integrate Google Gemini API

Description

The Google Gemini API is integrated with the FastAPI backend to provide AI-powered financial assistance. The backend sends the user's financial information, such as loan details, income, and expenses, to Gemini AI. The AI analyzes the data and generates personalized debt settlement recommendations, budgeting advice, repayment plans, and negotiation strategies. The generated responses are processed by the backend and displayed on the user's dashboard.

Milestone 3: Core Feature Development

This milestone focuses on developing the core functionalities of the **AI Powered Debt Relief & Financial Recovery Platform**. The main objective is to implement intelligent features that help users manage debts, analyze their financial health, generate AI-powered debt settlement recommendations, create negotiation strategies, and plan their finances effectively.

Activity 3.1: Develop the Loan Management Module

Description

The Loan Management module allows users to add, update, view, and delete loan information. Users can store details such as loan amount, lender name, interest rate, monthly EMI, due date, and repayment status. This module provides a centralized system for managing multiple loans efficiently.

Activity 3.2: Implement the Financial Health Analysis Module

Description

The Financial Health Analysis module evaluates the user's financial condition by analyzing income, expenses, debts, and repayment capacity. It calculates financial indicators such as debt-to-income ratio, monthly cash flow, and affordability, helping users understand their financial health and make informed decisions.

Code Blame 23 lines (19 loc) · 444 Bytes

```
1  from sqlalchemy import create_engine
2  from sqlalchemy.orm import declarative_base, sessionmaker
3
4  DATABASE_URL = "sqlite:///./debt_relief.db"
5
6  engine = create_engine(
7      DATABASE_URL,
8      connect_args={"check_same_thread": False}
9  )
10
11  SessionLocal = sessionmaker(
12      autocommit=False,
13      autoflush=False,
14      bind=engine
15  )
16
17  Base = declarative_base()
18  def get_db():
19      db = SessionLocal()
20      try:
21          yield db
22      finally:
23          db.close()
```

Activity 3.3: Develop the AI Debt Settlement Recommendation Module

Description

This module integrates **Google Gemini AI** to generate personalized debt settlement recommendations. Based on the user's financial information, the AI suggests suitable repayment options, settlement strategies, and customized plans to reduce financial burden and improve debt management.

Activity 3.4: Develop the AI Negotiation Strategy Module

Description

The AI Negotiation Strategy module helps users prepare professional negotiation plans for communicating with lenders. By analyzing financial data, Google Gemini AI generates personalized negotiation messages, repayment proposals, and settlement strategies that improve the chances of successful debt resolution.

Activity 3.5: Develop the Budget Planner and Financial

Recovery Module

Description

The Budget Planner module provides personalized monthly budgeting recommendations based on the user's income, expenses, and debt obligations. It also generates financial recovery plans, savings suggestions, and spending limits to help users improve their financial stability and achieve long-term debt recovery.

```
1  from fastapi import FastAPI
2  from database import engine, Base
3  from app.routes.loan_routes import router as loan_router
4  from app.routes.financial_routes import router as financial_router
5  from app.routes.settlement_routes import router as settlement_router
6  from app.routes.dashboard_routes import router as dashboard_router
7  from app.routes.strategy_routes import router as strategy_router
8  from fastapi.middleware.cors import CORSMiddleware
9  from app.routes.ai_routes import router as ai_router
10
11 from app.models.loan_model import Loan
12
13 app = FastAPI()
14 app.add_middleware(
15     CORSMiddleware,
16     allow_origins=["*"],
17     allow_credentials=False,
18     allow_methods=["*"],
19     allow_headers=["*"],
20 )
21 Base.metadata.create_all(bind=engine)
22
23 app.include_router(loan_router)
24 app.include_router(financial_router)
25 app.include_router(settlement_router)
26 app.include_router(dashboard_router)
27 app.include_router(strategy_router)
28 app.include_router(ai_router)
29
30 @app.get("/")
31 def home():
32     return {
```

Milestone 4: Frontend Development

This milestone focuses on developing a responsive, user-friendly, and interactive frontend for the **AI Powered Debt Relief & Financial Recovery Platform** using **React.js**. The frontend provides an intuitive interface that allows users to manage loans, analyze their financial health, receive AI-powered recommendations, and monitor their debt recovery progress. It communicates with the FastAPI backend through REST APIs and displays real-time information to users.

Activity 4.1: Design and Develop the User Interface

Description

The user interface is designed using **React.js**, HTML, and CSS to provide a modern and responsive experience. Pages such as Home, Login, Registration, Dashboard, Loan Management, Budget Planner, and AI Recommendation are created. The interface is optimized for desktop, tablet, and mobile devices to ensure a smooth user experience.



```
Code Blame 4 lines (4 loc) · 33 Bytes

1 node_modules
2 dist
3 .env
4 .DS_Store

Code Blame 13 lines (13 loc) · 518 Bytes

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>FinRelief AI | Debt Relief & Financial Recovery</title>
7 <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 100 100'%3E%3Ctext y='.9em' font-size='90'%3E%F0%9F%92%A0%3C/t
8 </head>
9 <body>
10 <div id="root"></div>
11 <script type="module" src="/src/main.jsx"></script>
12 </body>
13 </html>
```

Activity 4.2: Connect the Frontend with FastAPI Backend

Description

The React frontend is connected to the FastAPI backend using REST APIs. User actions such as registration, login, loan management, financial analysis, and AI recommendation requests are sent to the backend, and the responses are displayed dynamically without reloading the page.

```
Code Blame 21 lines (21 loc) · 408 Bytes
Raw [copy] [download] [edit] [diff]

1  {
2    "name": "finrelief-ai",
3    "private": true,
4    "version": "1.0.0",
5    "type": "module",
6    "scripts": {
7      "dev": "vite",
8      "build": "vite build",
9      "preview": "vite preview"
10   },
11   "dependencies": {
12     "axios": "^1.7.9",
13     "react": "^18.3.1",
14     "react-dom": "^18.3.1",
15     "react-router-dom": "^6.28.0"
16   },
17   "devDependencies": {
18     "@vitejs/plugin-react": "^4.3.4",
19     "vite": "^5.4.11"
20   }
21 }
```

Activity 4.3: Display AI Recommendations and Financial Reports

Description

The frontend displays AI-generated debt settlement recommendations, budgeting advice, financial health analysis, repayment schedules, and negotiation strategies in an interactive dashboard. Graphs, cards, and reports help users understand their financial condition and track their debt recovery progress.

Activity 4.4: Improve User Experience and Responsiveness

Description

The application is optimized to provide a smooth and responsive experience across different devices. Form validation, loading indicators, error messages, and intuitive navigation are implemented to improve usability. Responsive layouts ensure that the platform works efficiently on desktops, tablets, and smartphones.

```
1  {
2    "name": "finrelief-ai",
3    "version": "1.0.0",
4    "lockfileVersion": 3,
5    "requires": true,
6    "packages": {
7      "": {
8        "name": "finrelief-ai",
9        "version": "1.0.0",
10       "dependencies": {
11         "axios": "^1.7.9",
12         "react": "^18.3.1",
13         "react-dom": "^18.3.1",
14         "react-router-dom": "^6.28.0"
15       },
16       "devDependencies": {
17         "@vitejs/plugin-react": "^4.3.4",
18         "vite": "^5.4.11"
19       }
20     },
21     "node_modules/@babel/code-frame": {
22       "version": "7.29.7",
23       "resolved": "https://registry.npmjs.org/@babel/code-frame/-/code-frame-7.29.7.tgz",
24       "integrity": "sha512-Aup7aUOfpbAUg2ROOJN6Iw5f9DMB1zu0mIkM/malLQFN/YQgO48wCj0Kxa3sEHJvPVFg7s1R+qRInwXd2qhQKw==",
25       "dev": true,
26       "license": "MIT",
27       "dependencies": {
28         "@babel/helper-validator-identifier": "^7.29.7",
29         "js-tokens": "^4.0.0",
30         "picocolors": "^1.1.1"
31       },
32       "engines": {
33         "node": ">=6.9.0"
```

```

34     }
35   },
36   "node_modules/@babel/compat-data": {
37     "version": "7.29.7",
38     "resolved": "https://registry.npmjs.org/@babel/compat-data/-/compat-data-7.29.7.tgz",
39     "integrity": "sha512-locTkQyKvWIEgBzVrn8693ebc97F2U8ZHjbXwDXJ5Fn2TCpNwTlKcaKkLdHop5c/ic0FE7qt7Q9JC5hnKN6Gg==",
40     "dev": true,
41     "license": "MIT",
42     "engines": {
43       "node": ">=6.9.0"
44     }
45   },
46   "node_modules/@babel/core": {
47     "version": "7.29.7",
48     "resolved": "https://registry.npmjs.org/@babel/core/-/core-7.29.7.tgz",
49     "integrity": "sha512-RgHBCvtjb0K2gXSNBNIkNoEc9qoVEtau3hj8gEqKQuL3HZAibKarWFEI3Lfm6EYKkLa10h8eSrj9b+ch9H/VBA==",
50     "dev": true,
51     "license": "MIT",
52     "dependencies": {
53       "@babel/code-frame": "^7.29.7",
54       "@babel/generator": "^7.29.7",
55       "@babel/helper-compilation-targets": "^7.29.7",
56       "@babel/helper-module-transforms": "^7.29.7",
57       "@babel/helpers": "^7.29.7",
58       "@babel/parser": "^7.29.7",
59       "@babel/template": "^7.29.7",
60       "@babel/traverse": "^7.29.7",
61       "@babel/types": "^7.29.7",
62       "@jridgewell/remapping": "^2.3.5",
63       "convert-source-map": "^2.0.0",
64       "debug": "^4.1.0",
65       "gensync": "^1.0.0-beta.2",

```

```

65       gensync : ^1.0.0-beta.2 ,
66       "json5": "^2.2.3",
67       "semver": "^6.3.1"
68     },
69     "engines": {
70       "node": ">=6.9.0"
71     },
72     "funding": {
73       "type": "opencollective",
74       "url": "https://opencollective.com/babel"
75     }
76   },
77   "node_modules/@babel/generator": {
78     "version": "7.29.7",
79     "resolved": "https://registry.npmjs.org/@babel/generator/-/generator-7.29.7.tgz",
80     "integrity": "sha512-3kU1jHhIoWmU5hWboVao+qUEcyosB+5h8h3gKsW58YpxS1k/0BlAeV06SfPIpeh4eQv4yCPt1ayY14YFQ==",
81     "dev": true,
82     "license": "MIT",
83     "dependencies": {
84       "@babel/parser": "^7.29.7",
85       "@babel/types": "^7.29.7",
86       "@jridgewell/gen-mapping": "^0.3.12",
87       "@jridgewell/trace-mapping": "^0.3.28",
88       "jsesc": "^3.0.2"
89     },
90     "engines": {
91       "node": ">=6.9.0"
92     }
93   },
94   "node_modules/@babel/helper-compilation-targets": {
95     "version": "7.29.7",
96     "resolved": "https://registry.npmjs.org/@babel/helper-compilation-targets/-/helper-compilation-targets-7.29.7.tgz",
97     "integrity": "sha512-6h01Uq5kV9qlMXH80wUWcz43qP1PZv8T/LFRHvVf1B3eSuz3H0v1714Jz0l6KPC4e34q3+3Y0b0eijwQ==",

```

Milestone 5: Testing and Deployment

This milestone focuses on testing the **AI Powered Debt Relief & Financial Recovery Platform** to ensure that all modules function correctly and securely. After successful testing, the application is deployed so users can access it through a web browser. The deployment process ensures the platform is reliable, scalable, and ready for real-world use.

Activity 5.1: Application Testing

Description

Comprehensive testing is performed to verify that all modules work as expected. Functional testing ensures users can register, log in, manage loans, receive AI-powered debt recommendations, and access financial reports without errors. API testing verifies communication between the React frontend, FastAPI backend, SQLite database, and Google Gemini API.

Activity 5.2: Bug Fixing and Performance Optimization

Description

After testing, identified bugs and errors are fixed to improve the application's stability and performance. The system is optimized to provide faster API responses, efficient database operations, secure authentication, and a smooth user experience. Performance improvements ensure that the platform handles multiple user requests effectively.

Activity 5.3: Deploy the Application

Description

The completed application is deployed using **Render** for hosting. The source code is uploaded to **GitHub**, and the Render platform is configured to deploy both the frontend and backend services. Environment variables such as the Google Gemini API key are securely configured to ensure proper application functionality.

Activity 5.4: Verify the Deployed Application

Description

After deployment, the application is tested in the live environment. All major features—including user authentication, loan management, AI-generated recommendations, financial reports, and budgeting—are verified to ensure they work correctly. The deployed system is monitored for stability, security, and performance.

Code **Blame** 21 lines (17 loc) · 1.05 KB

```
1  MIT License
2
3  Copyright (c) 2026 SatyanarayanaVeeramallu
4
5  Permission is hereby granted, free of charge, to any person obtaining a copy
6  of this software and associated documentation files (the "Software"), to deal
7  in the Software without restriction, including without limitation the rights
8  to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
9  copies of the Software, and to permit persons to whom the Software is
10  furnished to do so, subject to the following conditions:
11
12  The above copyright notice and this permission notice shall be included in all
13  copies or substantial portions of the Software.
14
15  THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
16  IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
17  FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
18  AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
19  LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
20  OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
21  SOFTWARE.
```

Milestone 6: Project Completion and Documentation

This milestone focuses on completing the **AI Powered Debt Relief & Financial Recovery Platform** by preparing the final project documentation, organizing the source code, creating the GitHub repository, and demonstrating the application. The project deliverables are finalized and made ready for submission.

Activity 6.1: Prepare Project Documentation

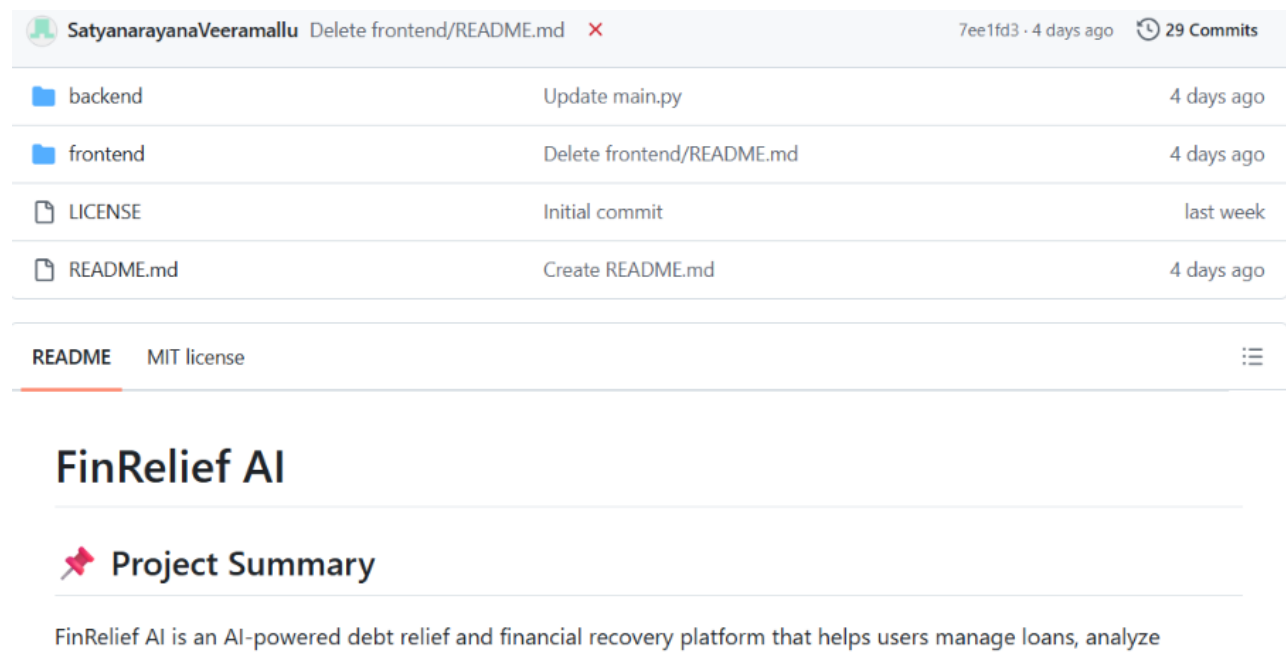
Description

The complete project documentation is prepared, including the project description, problem statement, system architecture, workflow, implementation details, module descriptions, testing results, screenshots, and conclusion. The documentation clearly explains the development process, technologies used, and the working of the AI Powered Debt Relief & Financial Recovery Platform.

Activity 6.2: Upload Source Code to GitHub

Description

The complete source code is organized and uploaded to a GitHub repository. All project files, including the React.js frontend, FastAPI backend, SQLite database scripts, and configuration files, are maintained using Git version control. This enables easy collaboration, version tracking, and future updates.



The screenshot shows a GitHub repository for 'FinRelief AI' by user 'SatyanarayanaVeeramallu'. The repository has 29 commits. The commit history table is as follows:

File	Commit Message	Time Ago
backend	Update main.py	4 days ago
frontend	Delete frontend/README.md	4 days ago
LICENSE	Initial commit	last week
README.md	Create README.md	4 days ago

The repository is licensed under MIT license. The project summary states: 'FinRelief AI is an AI-powered debt relief and financial recovery platform that helps users manage loans, analyze'.

Activity 6.3: Final Project Demonstration

Description

The completed application is demonstrated by showcasing its major features, including user registration, secure login, loan management, financial health analysis, AI-powered debt settlement recommendations, budgeting assistance, and financial recovery planning. The demonstration verifies that all modules function correctly and meet the project objectives.



[Dashboard](#)
[Add Loan](#)
[Loan Management](#)
[Analysis](#)
[AI Letter](#)
[Negotiation Strategy](#)

Dashboard

A live snapshot of your debt portfolio and financial health.

 timeout of 15000ms exceeded
 Retry


TOTAL LOANS
0
 Active loan accounts


TOTAL OUTSTANDING
₹0
 Sum across all loans


AVERAGE EMI
₹0
 Per loan, per month


FINANCIAL HEALTH
Review
 AI-assessed status

Financial Health Summary

No summary available yet. Add loans to generate insights.



Let's get started

You haven't added any loans yet. Add your first loan to unlock AI analysis.



[Dashboard](#)
[Add Loan](#)
[Loan Management](#)
[Analysis](#)
[AI Letter](#)
[Negotiation Strategy](#)

Loan Management

View, edit, and act on every loan in your portfolio.

 Add Loan

LOAN NAME	OUTSTANDING	EMI	MONTHLY INCOME	OVERDUE (MONTHS)	ACTIONS
education loan	₹26,000	₹15,000	₹30,000	2	Analyze Letter Strategy Edit Delete

Activity 6.4: Final Submission

Description

The final project deliverables are prepared and submitted. These include the source code, project documentation, presentation (PPT), project report, diagrams, screenshots, and deployment details. The project is reviewed to ensure that all objectives have been achieved successfully.

Negotiation Strategy

Select a loan to view its negotiation strategy.

Choose a loan to view its strategy:

education loan

Outstanding: ₹2,50,000

Project URL

<https://finrelief-ai.vercel.app/>

Video Demo Link:

https://drive.google.com/file/d/1w6xinPNoCLOB_8rhKKdwjmoDRmyEBCWU/view

Descriptions of Each Module

1. User Authentication Module

The User Authentication module provides secure registration and login functionality for users. It uses JWT (JSON Web Token) authentication to verify user identities and protect sensitive financial information. This module ensures that only authorized users can access their personal financial data and use the platform's services securely.

2. Loan Management Module

The Loan Management module allows users to add, update, view, and manage their loan details in one place. Users can store information such as loan amount, interest rate, lender name, EMI, due date, and repayment status. This module helps users organize multiple loans efficiently and monitor their repayment progress.

3. Financial Health Analysis Module

The Financial Health Analysis module evaluates the user's financial condition by analyzing income, expenses, savings, and outstanding debts. It calculates important financial indicators such as debt-to-income ratio, repayment capacity, and monthly cash flow. The analysis helps users understand their financial position and make informed financial decisions.

4. AI Debt Settlement Recommendation Module

This module uses Google Gemini AI to generate personalized debt settlement recommendations based on the user's financial information. It analyzes loan details, income, and expenses to suggest suitable repayment plans, settlement options, and financial strategies that help reduce debt and improve financial stability.

5. AI Negotiation Strategy Module

The AI Negotiation Strategy module generates intelligent negotiation suggestions for users who want to discuss repayment options with lenders. Using Google Gemini AI, the system creates personalized negotiation strategies, repayment proposals, and settlement messages that improve the chances of reaching mutually beneficial agreements.

6. Budget Planner Module

The Budget Planner module helps users manage their monthly income and expenses by providing personalized budgeting recommendations. It suggests spending limits, savings goals, and expense management strategies that enable users to control their finances and improve their overall financial health.

7. Financial Recovery Dashboard Module

The Financial Recovery Dashboard provides a complete overview of the user's financial status. It displays loan summaries, repayment progress, financial health reports, AI-generated recommendations, budgeting insights, and recovery status through interactive charts, graphs, and reports. This module allows users to track their progress toward becoming debt-free.

8. Report Generation Module

The Report Generation module creates detailed financial reports based on the user's loan information, repayment history, and AI recommendations. Users can view or download these reports to monitor financial progress, review repayment plans, and support better financial decision-making.

9. Database Management Module

The Database Management module uses SQLite with SQLAlchemy ORM to securely store and manage user accounts, loan details, financial records, AI-generated recommendations, and repayment history. It ensures efficient data retrieval, updates, and secure storage throughout the application.

10. AI Recommendation Engine

The AI Recommendation Engine is the core intelligence of the platform. It integrates with the Google Gemini API to analyze financial data and generate personalized debt management strategies, settlement recommendations, budgeting advice, financial recovery plans, and negotiation suggestions. This module enables users to receive real-time, AI-powered financial guidance.

Conclusion

The **AI Powered Debt Relief & Financial Recovery Platform** is an intelligent web application developed to help individuals effectively manage their debts and improve their financial well-being. By integrating **Google Gemini AI** with modern technologies such as **React.js, FastAPI, Python, SQLite, SQLAlchemy ORM, and JWT Authentication**, the platform provides personalized debt settlement recommendations, budgeting advice, financial health analysis, and AI-powered negotiation strategies.

The system enables users to organize loan information, monitor repayment progress, analyze their financial condition, and receive real-time guidance for better financial decision-making. Its secure, user-friendly, and responsive design makes debt management simple, efficient, and accessible.

Overall, the project demonstrates how Artificial Intelligence can transform traditional financial management by providing smart, personalized, and data-driven solutions. The **AI Powered Debt Relief & Financial Recovery Platform** empowers users to reduce financial stress, improve financial planning, and achieve long-term financial recovery, making it a valuable solution for modern personal finance management.

